

Magnetic Skyrmions

A $\text{\LaTeX}$ presentation template for Utrecht University

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Outline

Theory

Results

Background

$$H = H_J + H_D$$

Heisenberg exchange Hamiltonian : $H_J = -J \sum_r \vec{S}_r \cdot (\vec{S}_{r+\hat{x}} + \vec{S}_{r+\hat{y}}) - H_Z \sum_r \vec{S}_r$

Dzyaloshinskii-Moriya interaction Hamiltonian : $H_D = D \sum_r \vec{S}_r \times \vec{S}_{r+\hat{x}} \cdot \hat{x} + \vec{S}_r \times \vec{S}_{r+\hat{y}} \cdot \hat{y}$

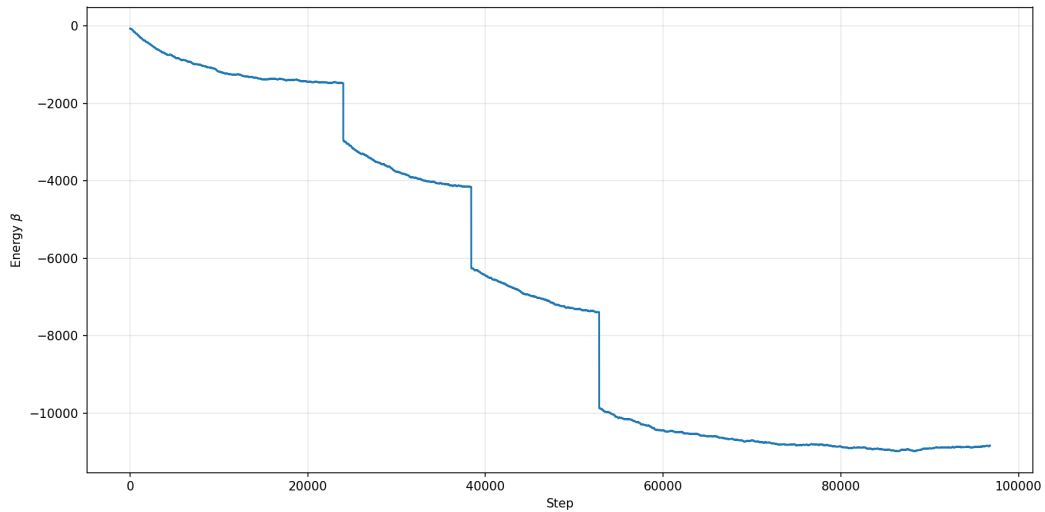
Simulation parameters

40 x 40 grid

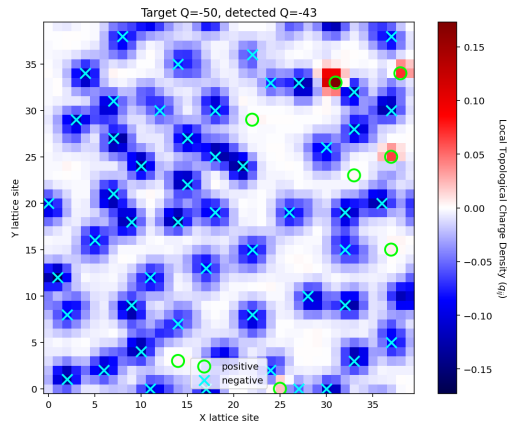
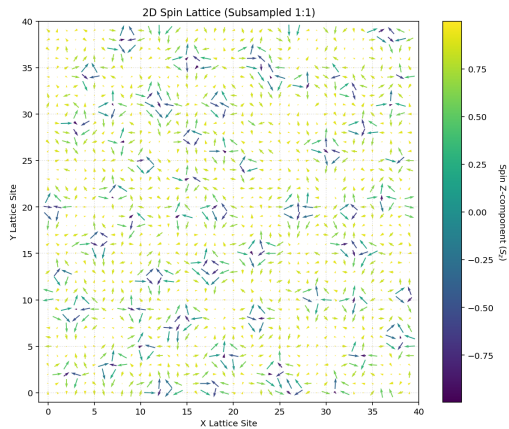
H, D being variable and J constant

beta being quenched

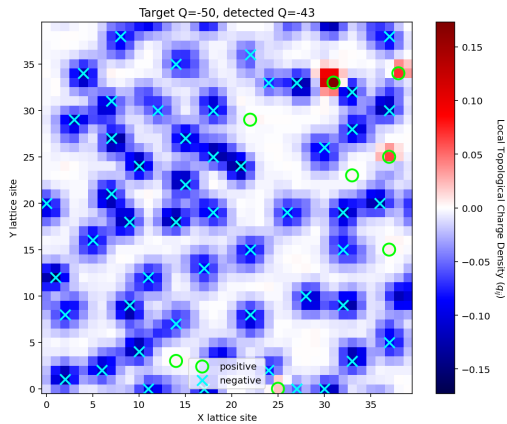
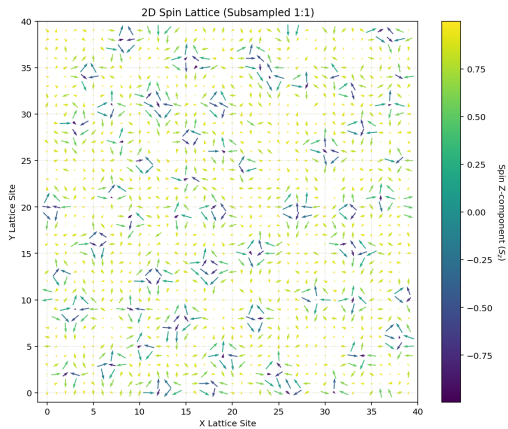
Quenching



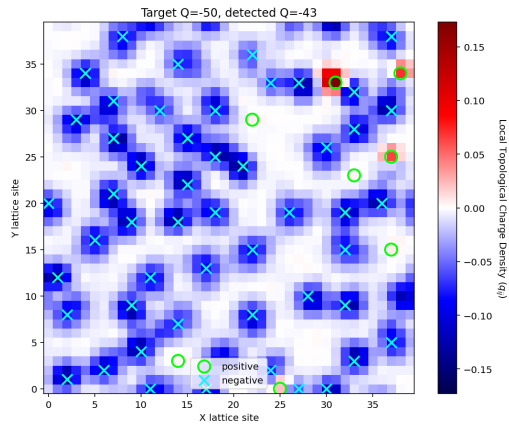
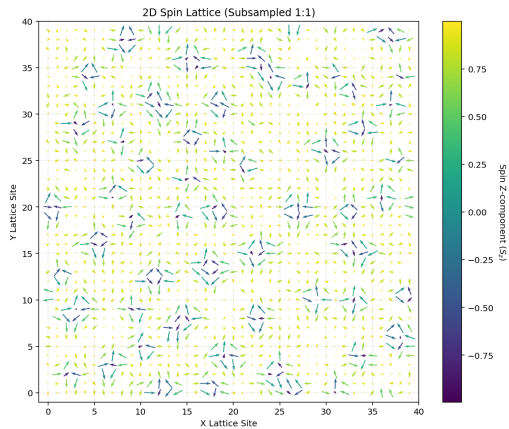
Spin lattice



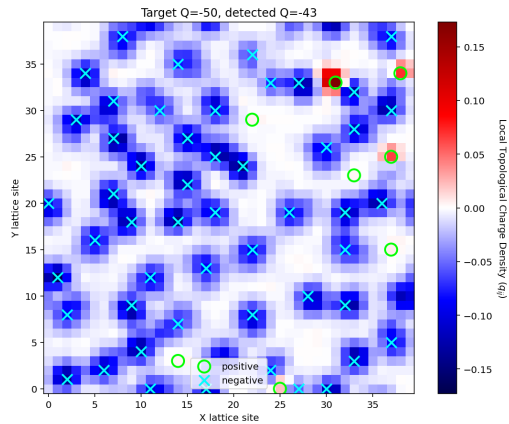
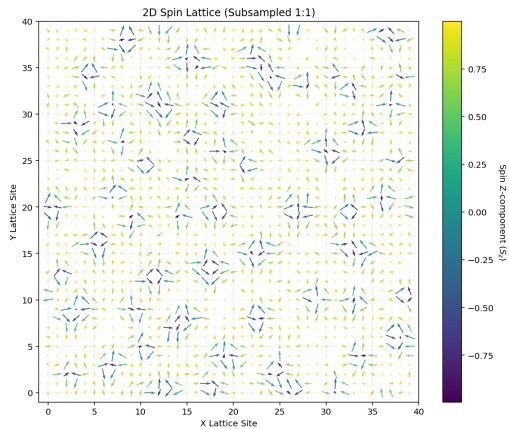
Wrapping number



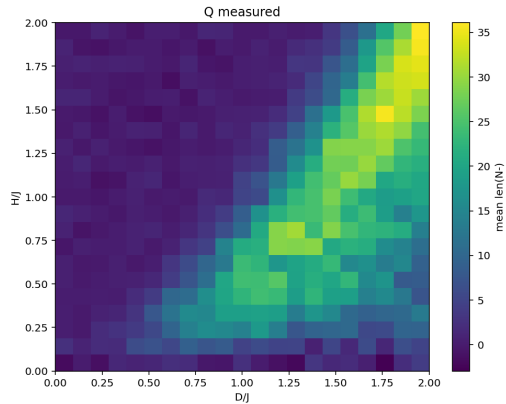
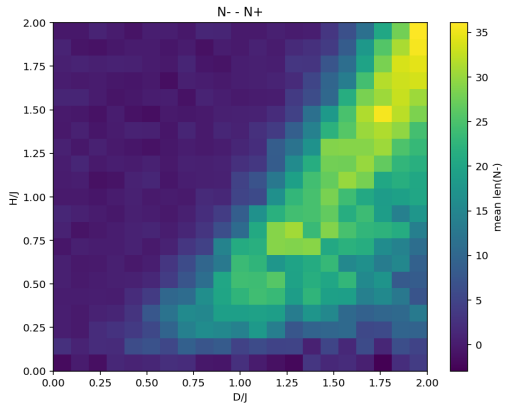
Skyrmion detection



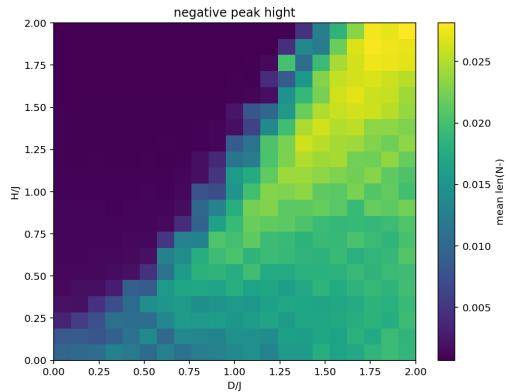
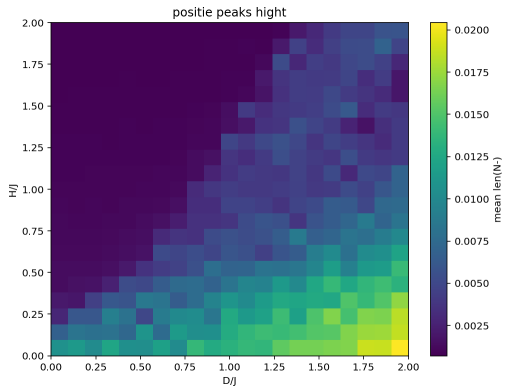
Skyrmion detection



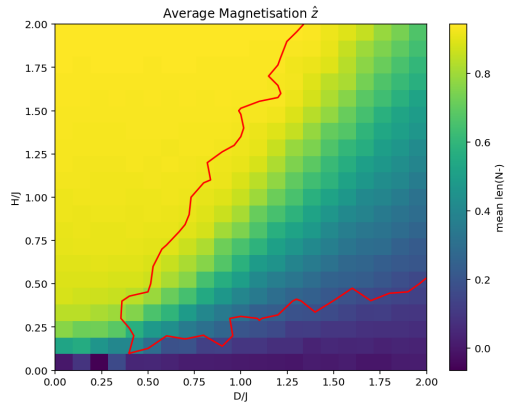
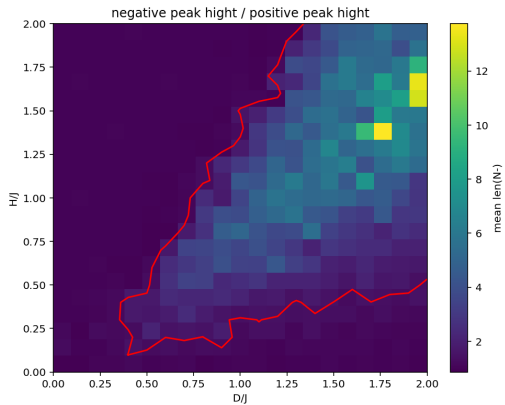
Phase Diagram



Phase Separation 1



Phase Separation 2



Temperature Dependence

